

8th Semester of B. Pharm. Examination
University Theory Examination May 2016
Pharmaceutical Technology-II (PH412)

Time:10:00 a.m to 1:00 p.m

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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| A | (i) Enlist formulation systems of pharmaceutical aerosols. Discuss any one in detail. | 05 |
| | (ii) Describe various components of continuous spray valve. | 05 |
| B | Discuss evaluation of a pharmaceutical suspension with emphasis on Zeta potential, Particle size and Rheological characteristics. | 10 |
| C | (i) Write a note on 'Dry Powder Inhaler'. | 05 |
| | (ii) State the merits and demerits of pharmaceutical aerosol as dosage form over other dosage forms. | 05 |

Q 2 Attempt any **TWO** of the following

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| A | (i) Explain the following tests for ‘Pharmaceutical aerosols’: | 04 |
| | (a) Flash Point (b) Vapour pressure | |
| | (c) Dosage with metered valve (d) Spray pattern | |
| | (ii) Describe various formulation additives of solution dosage form
and explain its evaluation parameters in detail. | 06 |
| B | Describe the apparatus and manufacturing methods for pharmaceutical aerosol in detail. | 10 |
| C | Discuss the formulation of a stable emulsion considering selection of oil,
emulsifier and phase volume ratio. Highlight the points to be considered
for different types of emulsion as per route of administration. | 10 |

SECTION- II

- Q 3** Attempt any **TWO** of the following
- A What do you mean by pilot plant? Explain the objectives, importance and steps to be taken for scale up of the pilot plant batch. 10
- B Explain in detail the formulation, preparation and packaging of following cosmetic products: 10
- (a) Lipstick (b) Nail Polish and Nail polish remover
- C (i) Define Parenteral preparation. Explain general consideration for formulation of parenteral formulations. Draw the layout of Parenteral manufacturing area. 05
- (ii) Describe the 'Air control' facility and explain the importance of it for the manufacturing of parenteral dosage forms. 05
- Q 4** Attempt any **TWO** of the following
- A Explain the following:
- (i) Cryoprotectant for parenteral dosage forms. 03
- (ii) Blow fill seal technology. 07
- B (i) Write a detailed note on: Skin care cosmetic products. 05
- (ii) Describe different tests on Plastic containers for formulation of parenteral dosage forms. 05
- C (i) Classify 'Ophthalmic products'. Explain formulation composition and preparation method of 'Eye drops'. 04
- (ii) Explain in detail: 'Bacterial Endotoxin test' and 'Pyrogen test' for parenteral dosage forms. 06
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